

WYNAU, Switzerland

WMO: 066430

Lat: 47.2550N

Lon: 7.7875E

Elev: 416

StdP: 96.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.5	(d) -12.0	(e) 1.4	(f) -7.5	(g) -9.3	(h) 1.8	(i) -5.1	(j) 8.7	(k) 1.4	(l) 7.7	(m) 2.7	(n) 2.6	(o) 60	(p) 0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.0	(c) 30.4	(d) 19.9	(e) 28.4	(f) 19.4	(g) 26.5	(h) 18.6	(i) 20.9	(j) 28.1	(k) 20.1	(l) 26.6	(m) 19.3	(n) 24.9	(o) 2.2	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.2	22.9	18.0	13.6	21.9	17.3	13.0	21.4	62.3	28.1	59.3	26.6	56.6	25.1	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 6.1	(c) 5.1	(e) -12.0	(f) 33.1	(g) 2.9	(h) 1.7	(i) -14.0	(j) 34.3	(k) -15.7	(l) 35.3	(m) -17.3	(n) 36.3	(o) -19.4	(p) 37.5		
(a) 7.2	(b) 6.1	(c) 5.1	(e) -12.4	(f) 22.8	(g) 2.9	(h) 1.0	(i) -14.4	(j) 23.5	(k) -16.1	(l) 24.1	(m) -17.7	(n) 24.6	(o) -19.8	(p) 25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	0.6	1.6	5.6	9.4	13.8	17.7	19.3	18.9	14.7	10.5	4.8	1.4		
	DBStd	7.55	3.78	3.96	3.67	3.53	3.31	3.57	3.07	3.07	3.13	3.50	3.49	3.58		
	HDD10.0	1191	291	235	142	53	7	1	0	0	2	37	158	266		
	HDD18.3	3224	549	468	396	268	144	55	25	31	115	244	406	524		
	CDD10.0	1155	0	0	4	36	124	230	289	275	143	52	2	0		
	CDD18.3	148	0	0	0	0	4	35	56	47	6	0	0	0		
	CDH23.3	1361	0	0	0	7	68	343	502	392	48	1	0	0		
	CDH26.7	388	0	0	0	1	10	100	152	122	4	0	0	0		
(14) Wind	WSAvg	1.7	2.0	2.0	2.0	1.8	1.7	1.5	1.4	1.2	1.3	1.4	1.6	1.9		
(15) Precipitation	PrecAvg	1093	73	69	70	82	106	111	117	110	88	87	88	93		
	PrecMax	1383	187	168	248	179	183	230	204	213	157	163	211	199		
	PrecMin	752	16	23	24	9	34	26	37	17	42	11	4	0		
	PrecStd	163	44	37	49	45	44	47	37	46	36	41	50	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	13.6	19.1	24.2	28.0	32.2	33.0	33.0	27.2	21.3	15.5	10.8		
		MCWB	8.5	8.2	11.0	14.2	17.8	20.8	20.6	20.7	18.8	15.6	11.6	9.0		
	2%	DB	9.3	10.9	16.7	21.6	25.1	29.5	30.3	29.7	24.7	19.0	12.5	9.2		
		MCWB	7.2	7.2	10.0	13.0	16.3	19.6	19.9	19.9	17.8	14.8	10.2	7.8		
	5%	DB	7.5	9.0	14.3	19.3	23.1	27.2	28.2	27.5	22.4	17.1	10.9	7.9		
		MCWB	5.9	6.1	9.0	11.9	15.4	18.9	19.4	19.4	16.8	13.8	9.1	6.7		
	10%	DB	5.9	7.2	12.0	16.9	20.8	25.0	26.1	25.2	20.2	15.3	9.6	6.6		
		MCWB	4.6	5.0	8.0	10.7	14.3	17.9	18.4	18.5	15.7	13.1	8.2	5.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	9.1	11.8	15.0	19.0	22.1	21.9	21.9	19.9	16.4	12.2	9.5		
		MCDB	10.7	12.4	17.4	23.3	26.0	29.6	29.8	28.9	25.3	19.7	14.3	10.4		
	2%	WB	7.5	7.6	10.6	13.5	17.3	20.5	20.7	20.8	18.5	15.4	10.7	8.1		
		MCDB	9.0	10.5	15.2	20.3	23.3	27.5	27.9	27.7	23.2	18.1	12.3	9.1		
	5%	WB	6.0	6.4	9.5	12.4	16.1	19.4	19.9	20.0	17.4	14.4	9.4	6.8		
		MCDB	7.3	8.8	13.3	18.2	21.6	25.6	26.4	26.0	21.3	16.5	10.7	7.8		
	10%	WB	4.7	5.2	8.4	11.3	14.9	18.4	19.1	19.1	16.4	13.3	8.3	5.6		
		MCDB	5.8	6.9	11.7	15.8	19.5	23.8	24.6	24.1	19.5	15.1	9.4	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	7.3	9.8	11.0	10.6	10.9	11.0	10.6	9.7	7.7	5.5	4.4		
		MCDBR	7.1	11.0	14.9	16.3	15.3	15.2	15.0	14.9	13.1	10.2	7.8	6.0		
	5% WB	MCWBR	5.3	7.2	8.4	8.2	7.2	6.3	5.7	5.9	6.6	6.0	5.4	4.7		
		MCWBR	6.5	10.2	13.2	15.0	13.6	13.8	13.4	13.2	11.9	9.5	7.4	5.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.294	0.319	0.353	0.388	0.400	0.403	0.405	0.397	0.374	0.351	0.317	0.291		
		taud	2.446	2.380	2.339	2.275	2.287	2.318	2.308	2.344	2.388	2.440	2.478	2.457		
	Ebn at Noon	Ebn at Noon	799	846	865	864	865	862	855	847	833	797	761	757		
		Edh at Noon	66	87	106	124	127	124	124	115	100	82	65	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.18	2.02	3.23	4.35	5.01	5.88	5.68	4.94	3.76	2.38	1.30	0.96		
	RadStd	RadStd	0.15	0.30	0.48	0.65	0.49	0.54	0.51	0.47	0.36	0.24	0.16	0.17		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only	Station Only	N/A	N/A	N/A	N/A	+0.54	+0.52	N/A	N/A
(47) Regional (118 neighbors)	Regional (118 neighbors)	N/A	N/A	N/A	+0.74	+0.41	+0.31	N/A	N/A

Nomenclature: See separate page